8-1 Computer Networks

Point!

1 Information and Communication Networks

(1) ([1]**Information and communication network**) : A network resembling a web that utilizes communication lines to transmit information.

(2) ([2]**Computer network**) : A communication system that connects information communication devices like computers and smartphones, enabling them to exchange data with one another.

(3) Types of Networks

[1] ([3]**LAN**) : A network that connects devices within a limited area, such as within a school or home.

- ([4]**Wired LAN**) : A method of exchanging data by directly connecting to cables.
- ([5]**Wireless LAN**) : A method of exchanging data via wireless radio waves instead of cables.

[2] ([6]**WAN**) : A network that connects a broader area.

(4) ([7]**Internet**) : A network that connects LANs and WANs on a global scale.

(5) Connecting to the Internet

[1] ([8]**ISP (provider)**) : A company that facilitates connection to the internet. It is necessary to sign a contract to connect to the internet.

[2] ([9]**Router**) : A device that relays and forwards data between different networks.

[3] ([10]**Hub**) : A central device that connects multiple cables.

2 Network Usage Patterns

(1) Server and Client

[1] ([11]**Server**) : A computer that provides various services.

- ([12]**File server**) : A server that manages files.
- ([13]**Printer server**) : A server that handles print processing.
- ([14]**Mail server**) : A server that handles the sending and receiving of emails.
- ([15]**Proxy server**) : A server that acts as an intermediary for accessing the internet.

[2] ([16]**Client**) : A computer that requests various services from a server.

(2) Network Usage Patterns

[1] ([17]**Client-server system**) : A method where services are provided and received between a server and a client.

[2] ([18]**Peer-to-peer system**) : A method where computers are in an equal relationship, providing and receiving services to and from each other.

3 Communication Methods

(1) ([19]**Circuit switching method**) : A communication method where a communication path is established between the sender and the recipient before starting the transmission to exchange information.

(2) ([20]**Packet**) : A unit for dividing data into small chunks when transmitting data over a network.

(3) ([21]**Packet switching method**) : A communication method that involves dividing the data being transmitted into packets for exchange.

(4) Advantages and Disadvantages of Communication Methods

	Circuit switching method	Packet switching method
Advantages	Once a communication path is established, stable communication can be achieved.	Even if the network is congested, data can be sent gradually, so the possibility of being completely unable to communicate is low.
Disadvantages	The communication path is occupied by the user, so it is not possible to use the line simultaneously with others.	Packets can sometimes be lost along the way or be affected by network congestion.

Warm Up

Answer the following questions.

(1) Choose one statement that is incorrect regarding network construction devices from the options **A** to **D** below, and answer using the letter.

A Organizations or companies that facilitate connections to the internet are referred to as providers.

B A network limited to areas such as within a school or a home is referred to as a WAN.

C A device that gathers and connects multiple cables is referred to as a hub.

D A device that relays and transfers between different networks is referred to as a router.

(2) Choose the statement that best fits into the blanks [1] to [4] in the following table from the options **A** to **D** below, and answer using the letters.

	Advantages	Disadvantages
Circuit switching method	[1]	[2]
Packet switching method	[3]	[4]

A The line is occupied, which ensures stable communication.

B There may be delays or loss of data during transmission.

C Only the number of lines necessary for communication is required.

D It is possible to handle many communications simultaneously.

Explanation

(1) A network limited to areas such as within a school or a home is referred to as a LAN. Therefore, **B**

(2) [1] **A** [2] **C** [3] **D** [4] **B**

Try

1 Choose the term that best fits into the blanks [1] to [6] in the following sentences from the options **A** to **H** below, and answer using the letters.

> A network within a relatively limited area, such as within a school or home, is referred to as a ([1]), while a network that covers a wider area is referred to as a ([2]). A network that connects ([1]) and ([2]) on a global scale is called the ([3]). In order for regular households to use it, they need to have a contract with a business that facilitates connection to the internet, known as a ([4]).
>
> Additionally, computers within a local range are typically connected via cables through a concentrator device known as a ([5]). When connecting different networks, an intermediary device called a ([6]) is used.

A LAN **B** WAN **C** Wi-Fi **D** Hub
E Server **F** Router **G** Internet **H** ISP (provider)

2 Answer the following questions.

(1) Choose the term that best fits into the blank in the following sentence from the options **A** to **D** below, and answer using the letter.

> A system configuration in which computers connected through a network are structured to allocate roles as service providers and service receivers is referred to as ( ).

A Client-server system **B** Peer-to-peer system
C Local Area Network (LAN) **D** Wide Area Network (WAN)

(2) Fill in the blanks with the appropriate words.

[1] A server that acts as an intermediary for accessing the internet is referred to as a ( ).
[2] A server that manages files is referred to as a ( ).
[3] A computer that requests various services from a server is referred to as a ( ).

3 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> There are ([1]) methods and ([2]) methods in communication systems. The ([1]) method is a communication method where a communication path is established between the sender and the recipient before starting communication to exchange information. The ([2]) method is a communication method where the data to be sent is divided into small units called packets for exchange.

(2) Choose all appropriate statements that describe circuit switching method and packet switching method from the options **A** to **D** below, and answer using the letters.

A Circuit switching method allows for high communication speeds because the communication path is shared with other users.
B In circuit switching method, once a connection is established, stable communication is possible because the communication path is occupied.
C Packet switching method involves sharing the communication path, which can result in lower communication speeds.
D Packet switching method has a low likelihood of complete communication failure, even when the network is congested.

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of a WAN from the options **A** to **D** below, and answer using the letter.

A It can only be connected within limited buildings such as schools, companies, and homes.

B It is possible to connect areas that are far apart.

C It is a communication network that connects information and communication devices, such as computers, to exchange information with each other.

D It is possible to interconnect networks around the world.

(2) Choose the term that best fits into the blanks in the following sentences from the options **A** to **D** below, and answer using the letters.

[1] The method of exchanging data by directly connecting cables is referred to as ( ).

[2] The method of exchanging data using wireless radio waves instead of cables is referred to as ( ).

[3] The device that relays and forwards data between different networks is referred to as a ( ).

A Wired LAN **B** Router **C** Hub **D** Wireless LAN

2 Answer the following questions.

(1) Choose one statement that is incorrect regarding the usage patterns of networks from the options **A** to **D** below, and answer using the letter.

A A server that manages data and files on a network is referred to as a file server.

B A computer that requests various services from a server is referred to as an ISP (provider).

C A server that accesses the internet on behalf of a user is referred to as a proxy server.

D A method where computers are in an equal relationship, providing and receiving services to and from each other, is referred to as a peer-to-peer system.

(2) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

- A server that handles printing processes such as copying and printing is referred to as a ([1]).
- A server that facilitates the sending and receiving of emails between computers is referred to as a ([2]).
- The usage pattern where a server and a client connect over a network and exchange services is referred to as ([3]).

A Client-server system **B** Mail server

C Internet **D** Printer server

(3) If the following statements [1] to [4] pertain to circuit switching method, answer with “A.” If the statements pertain to packet switching method, answer with “B.”

[1] Once a communication path is established, stable communication can be achieved.

[2] Since it can send data in small portions, the possibility of being unable to communicate is low, even if the network gets congested.

[3] Delays, tampering, or loss may occur during data transmission.

[4] Since you cannot use the line simultaneously with others, there is a need for a number of lines dedicated to communication.